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# GD32 LCD æ‰œ«èj”ç•Eé¢é…¢½@ë®á½•## 1. ç»æ ‡ äœ”â½“â‰œ GD32F303RCT6  
ã¼€ä¢•æ¢¿çš,, LCD â±¤â¼•â„Šæ³¼çºŒ¹¼š ``text Jasper NB `` è|æ±±  
,ä»¢ç„¶,¶ä¿•æE¢âŽŸăŽ, `GD32F303RxT6 KIT` ç»@â½•â¢âª»î¼(Eä, ¢ä¿@æ””âŽŸăŽ,æ-  
‡ä»¶æ€, ## 2. ä½¿ç””çš,, LCD çº³¼• æœæä^¢ä½¿ç”” KIT ä,çš,, 1.14  
â¬,çº³¼•î¼(Eä¢Žç»¢â®žæµ«ç¿@ë®ðä,° 1.47 â¬,â±¤î¼(Eä•²â¬‡æ¢¢çä°î¼š ``text  
GD32F303RxT6 KIT/3. SDK/GD32F303RCT6 Demo/Examples/LCD 1.47inch ``  
æ ,ä¿fê©±âš”æ¬‡ä»¶î¼š ``text Components/Devices/TFTLCD/tftlcd.c Components/Devices/  
TFTLCD/tftlcd.h Components/Devices/TFTLCD/tft_font.h `` è¿™äº»æ—  
‡ä»¶ä»¢ç„¶,¶ä¿•ç•™äœ”âŽŸăŽ, KIT ä,î¼(Eæœ¬â•¥ç”¢äªäœ”`firmware/CMakeLists.txt` ä,-  
â¼•ç””â®fä»¬î¼(Eä, ¢ç”æŽŸä¿@æ””âæ€, ## 3. LCD æŽŸçº¿â®Eä¼•è„š æ¹æ¢® KIT çš,, `LCD  
1.14inch` çº³¼•î¼(ELCD ä½¿ç”” SPI2 â®E GPIO æŽŸšâ®¶„šä€, | âšŸÿè½ | GD32 â¼•è„š | ---  
| --- || SPI SCK | PB3 || SPI MOSI | PB5 || LCD DC | PB4 || LCD RST | PB6 || LCD PWR / BL  
| PB7 || è¾¼...âš©æŽŸšâ®¶„š | PA15 | æ³”æ„,¢î¼š - `PB3`ä€¢`PB4`ä€¢`PA15` ä,Ž JTAG  
â¢¢ç””ç»,ä...³ä€, - â•¥ç”¢ä,â•²ç»¢¢èºfç””  
`gpio_pin_remap_config(GPIO_SWJ_SWDPENABLE_REMAP, ENABLE)`î¼(Eä...³é—  
JTAGä€¢ä¿•ç•™ SWDä€, - è¿™æ¬æ¬æ—çèf½ä½¿ç”” LCD ä¼•è„š,šî¼(Eä¹Ÿä, ¢ä½¿šä...³é— ST-  
Link çš,, SWD çfšâ½•èf½âš”ä€, ## 4. â•¥ç”¢æ””âš” æœ¬æ¬¡æ””âš”â...éf”â¢ç””Ÿäœ”  
`firmware/` ç»@â½•ä€, æ¬âçèžæ¬‡ä»¶î¼š ``text firmware/include/main.h firmware/include/spi.h  
firmware/src/lcd_spi.c `` ä¿@æ””æ¬‡ä»¶î¼š ``text firmware/CMakeLists.txt firmware/include/  
gd32f30x_libopt.h firmware/include/systick.h firmware/src/systick.c firmware/src/main.c `` ##  
5. SPI2 é€,é...¢ `firmware/src/lcd_spi.c` â®žçŽ”º† KIT LCD  
é©±âš”éœœè|¢çš,,ä,ðä,â¿½¿æ•ºî¼š ``c void spi_lcd_init(void); void SPI2_WriteBytes(uint8_t  
*pbuffer, uint32_t length); `` é...¢ç½®â®,æ•ºî¼š ``text SPIî¼šSPI2 æ¬¡â¼•¢î¼šMaster  
â¼•è¾¼“æ¬â¢`î¼šâ¢•çº¿â¢`é€¢ ä,šâðšâ¢¢î¼š8 bit æž¢æ€/š/  
ç»,ä½¿¢î¼šSPI_CK_PL_HIGH_PH_2EDGE NSSî¼šè½ ä»¶æŽŸšâ®¶„š éç„â¬†éç”î¼šSPI_PSC_8 â-  
—èš,â¢¢î¼šMSB first `` ## 6. CMake æŽŸŸä...Ÿæ¬â¼•¢ `firmware/CMakeLists.txt` âçžâš  
LCD é©±âš”è¬â¼„,î¼š ``cmake set(LCD_ROOT "${PROJECT1_DIR}/GD32F303RxT6 KIT/  
3. SDK/GD32F303RCT6 Demo/Examples/LCD 1.14inch/Components/Devices/TFTLCD") ``  
âçžâš æº¢¢ ¢î¼š ``cmake src/lcd_spi.c "${LCD_ROOT}/tftlcd.c" "${PERIPH_ROOT}/  
Source/gd32f30x_spi.c" `` âçžâš â¢æ¬‡ä»¶è¬â¼„,î¼š ``cmake "${LCD_ROOT}" ``  
âçžâš â¢è®¼â¢æ¬‡ä»¶î¼š ``c #include "gd32f30x_spi.h" `` ## 7. æ³¼çº³ä»£ç ¢ `firmware/  
src/main.c` ä,âš ä...Ÿî¼š ``c #include "spi.h" #include "tftlcd.h" `` âªâšâCEæµ¢ç”î¼š ``c  
systick_config(); board_init(); spi_lcd_init(); LCD_Init(); `` æ³¼çº³â¬†...â®î¼š ``c  
LCD_Clear(BLACK); BACK_COLOR = BLACK; POINT_COLOR = CYAN;  
LCD_ShowString(48U, 50U, 160U, 32U, 32U, "Jasper NB"); `` æ³¼çº³æ¬æœï¼š -  
èf(Eæ™î¼šé)“è‰º² - â—â½“èçœè‰º²î¼šé”è‰º² - â—â½“â®šâ®¢î¼š32 - ä½¿¢½®î¼š1.14  
â¬,æªâ±¤ä,æŽŸŸè¿;â±...ä, ## 8. æž„â® æ‰œšè¿Eî¼š ``sh cd ... cmake -S firmware -B  
firmware/build -G Ninja -DCMAKE_TOOLCHAIN_FILE=../firmware/cmake/arm-none-eabi-  
gcc.cmake -DCMAKE_BUILD_TYPE=Release cmake --build firmware/build ``  
æœ¬æ¬¡æž„â®ºç»“æœœï¼š ``text FLASH: 25484 B / 256 KiB RAM: 4240 B / 48 KiB `` è¬  
`æŽŸî¼š - LCD â—â”â®Eç»~â³¼â¿½¿æ•ºâš ä...Ÿâ¢Žî¼(Flash â¢ç””ä»Žçº¿ 3 KiB âçžâš âºçº¿  
25 KiBä€, - RAM â¢ç””âçžâš âºçº¿ 4.2 KiBi¼(Eä,»è|â¢E...â¢« LCD â¢`é€¢¢ç¼“â¬†âCEäœ  
, - âŽŸăŽ, Keil â•¥ç”¢â£æ¬Ž`IRAM(0x20000000,0x0C000)`î¼(Eæ‰œ  
€ä»Ÿâ½“â‰œ“é“¼æŽŸŸè„š,šæœ¬æE‰œ 48 KiB SRAM ë®¼ç½®âœ, æž„â®æ—  
¶â¿çŽ”º†âŽŸăŽ, LCD æ¬‡ä»¶¶çš,,èŸä¹² warningî¼(Eä¼•ä¿,âðšè¿Eæ””é‡šä€¢char ä,æ ‡ç‰œäœ  
,è¿™äº»æ¢Ÿè‡ª KIT âŽŸăŽâšé©±âš”æ¬  
‡ä»¶î¼(Eä, ¢â½±â“¢æœ¬æ¬¡â®ä»¶ç”Ÿæ¬â®Eçfšâ½•ä€, ## 9. çfšâ½• æŽŸçæµ« ST-Linkî¼š  
``sh st-info --probe `` â®žé™...ç»“æœœï¼š ``text Found 1 stlink programmers version:  
V2J37S7 flash: 262144 (pagesize: 2048) sram: 65536 chipid: 0x414 dev-type: F1xx_HD ``  
æ³”æ„,¢î¼š`st-info` éçšè¿¿ STM32F1 ä...¼â®¹â¢¢ë®@æšŸâš”`sram:  
65536î¼(Eä½¿èŸæ¼¼ä, ¢èf½â½¿œä,ºæœ¬æ¢¿è“¼æŽŸŸè„š,šæœ¬â¾¼æ¢®äœ  
,æœ¬æ¢¿äŽŸăŽ, Keil â•¥ç”¢ä½¿ç”” 48 KiB SRAMî¼(Eäâ®žé™...  
èºfè¬â¹Ÿç¿@ë®ðä `é¿¶æ””¼âº`0x20010000` ä¼šèš¿â¢æœ”çº¿è”™è—ä€, çfšâ½•î¼š ``sh st-
```

flash --reset write firmware/build/gd32f303rct6_firmware.bin 0x08000000 `` `@žé™...
ç»“æžœi¼š ``text Attempting to write 25244 (0x629c) bytes to stm32 address: 134217728
(0x8000000) 13/13 pages written Flash written and verified! jolly good! NRST is not connected
--> using software reset via AIRCR `` `æŽi¼š - à°ä»¶ā²ā†™ā...¥ Flash èµ·āš<āœ°ā◆
`0x08000000`ā€, - `st-flash` è†°āš”æ jé°CEæ°◆āšŸā€, - ā½°“ā%◆æœ°èžžæŽ¥ NRSTi¼CEā æ-
œā½žç””è½°ā»¶āœ◆ā½◆i¼CEē™ā,◆æ°ē”™è°ā€, ## 10. è°»āžæ jé°CE è°»āž Flashi¼š ``sh
st-flash read firmware/build/gd32f303rct6_firmware_readback.bin 0x08000000 25484 ``
æ”è¼°fæœ°āœ°ā°ā»¶ā,Žè°»āžā°ā»¶i¼š ``sh cmp firmware/build/gd32f303rct6_firmware.bin
firmware/build/gd32f303rct6_firmware_readback.bin `` ç»“æžœi¼š ``text æ
— è¼°“ā†°i¼CEèj°çœ°ā@CEā...“ā,€è†° ``SHA256i¼š ``text
fd4b70e163f8b8826224a1d9addb642d04829d9938f04d1fdf54921c56dfe4eb firmware/build/
gd32f303rct6_firmware.bin
fd4b70e163f8b8826224a1d9addb642d04829d9938f04d1fdf54921c56dfe4eb firmware/build/
gd32f303rct6_firmware_readback.bin `` ## 11. SRAM æ°éj¶äž@æfä,ŽèžçèjCEé°CE°◆ LCD
ç%°æœ°æœ°ā°◆çfšā½°ā◆Žæ²jæœ%œžç>ā...¥ `main()`i¼CEOpenOCD æ°¼çœ°āœ
„ç◆†ā™”ā°œāœ”`HardFault\_Handler`ā€, ā...³é”@ā°,ā°ā™”i¼š ``text CFSR = 0x00009600
HFSR = 0x40000000 BFAR = 0x2000fffc MSP = 0x2000ffd8 `` ā◆«ä¹%œi¼šā¼,ā,ā◆“ç”Ÿæ—
¶è@žè—@ā°†`0x2000fffc`i¼CEèj°æ”āžŸé“¼æŽ¥è„šæœ° 64 KiB SRAM é...
◆ç½°@,çš„æ°éj¶é™„èç°āœ°ā◆æfä,āžŸāž, Keil ā°Ÿç°<ā@žé™...āf°æ°Ži¼š ``text
IRAM(0x20000000,0x0C000) `` āiŸā°±æ° 48 KiB SRAMā€,ā æ-
œā²ā°†é“¼æŽ¥è„šæœ°æ”ā,°i¼š ``ld RAM (xrw) : ORIGIN = 0x20000000, LENGTH = 48K
`` äž@æfā°Ž`\_estack = 0x2000c000`ā€,OpenOCD é°CE°◆ç»“æžœi¼š ``text pc =
0x08000130 // main msp = 0x2000c000 CFSR = 0x00000000 HFSR = 0x00000000 `` ç»šç»
èçèjCE 1.5 çš°ā◆Žā°†æ°jæš,ā°œi¼CEā°◆æœ°èç>ā...¥ HardFaulti¼CE`CFSR/HFSR`
äž@æCE°ā,° 0ā€, ## 12. ā½°“ā%◆ç»“è@° æœ°æ°jā²ç»“ā@CEæ°◆i¼š - æŽ¥ā...¥ KIT çš„ 1.14
ā°, LCD é©±āš”ā€, - é...◆ç½°@ SPI2 ā°CE LCD æŽšā°¶ GPIOā€, - ā°ā»¶ā◆āš”ā°Žæ%œšèjCE
LCD ā°āš<āCE—ā€, - ā±◆ā¹·æ„...é°“ā°ā€, - æ°¼çœ°é°“è%² `Jasper NB`ā€, -
ā°ā»¶æž„ā°æ°◆āšŸā€, - ST-Link çfšā½°ā°āšŸā€, - Flash è°»āžæ jé°CEā,€è†°ā€, - ā²äž@æ
£ SRAM é...◆ç½°@ā°¼è†°çš„æ—©æœ°Ÿ HardFaultā€, ## 13. ā,æžœā±◆ā¹·æ²jæœ%œæ°¼çœ°
ā¼°ā...`æfææŸŸi¼š 1. ā¼°ā◆æ°çžā@žé™...ā@%œè£...çš„ā±◆ā¹·æ°ā◆ä,° 1.14 ā°,ā€, 2.
ā±◆ā¹·æŽ°ç°çjæ°æ°ā°šæ°ā°æžŸèšjè%°āŸi¼š, 3. ST-Link çfšā½°ā°āžæ°çjā°æ°ā°jæ-
£ā,āœ°ā½°ā€, 4. `PB3/PB5/PB4/PB6/PB7/PA15` æ°ā°èç«ā...¶ā—āœ—è@¼ā◆ç””ā€, 5.
è°Ÿā@žé™...ā±◆ā¹·ā,° 0.96 ā°,æ°— 1.47 ā°,i¼CEœœèè|ā°†æ°çā° KIT ā°ā°” LCD
çœ°ā¼<ç>@ā½°ā€, ā◆é€%œ LCD çœ°ā¼<ç>@ā½°·i¼š ``text Examples/LCD 0.96inch BOE
Examples/LCD 1.14inch Examples/LCD 1.47inch `` ā½°“ā%◆ā°Ÿç°<é°“è@œā½žç””i¼š ``text
Examples/LCD 1.14inch `` ## 14. æ%œ<èj°éžžæ¼ç°CEé°çæ°æ°° ā²ā°†āžŸæ°Ÿçš„, LCD
éçœè%²ā¼°çŽ°è Šæ°ç°CEé°çæ°çæ°çæ°°œ%œ<èj°æ%œ<çŽ°éžžæ¼ā„ç°CEé°çæ°æ°, æ°¼çœ°ā°†...
ā@i¼š - éj¶é°æ°¼çœ°`HELLO JASPER`ā€, - èfCEæ™”ā½žç””è“◆è°“è%²ā€, - ä,é—
`æ°¼çœ°āœšā°æ—¶è—`HH:MM` ā°CEçš°æ°ā€, - ā°é°æ°¼çœ°æ°Ÿæœ°Ÿç¼°ā°†™ā°CEæ—
Ÿæœ°Ÿi¼CEā¼°ā, `FRI 2026-05-15`ā€, - ä,◆ā°†°ā...“ā±◆ā¼°çŽ°ā°æ—
°ā,◆ā°CEèfCEæ™”è%²i¼CEā°æ°◆çš°ā±€é°æ°æ°æ—¶è—`āCEāŸŸā€, ā½°“ā%◆æ—¶è—
`æ°Ÿæ°◆i¼š - ā°ā»¶ā◆āš”æ—¶ā°Žç¼°è°ā@°`_DATE_` ā°CE`_TIME_` è°»ā◆
æž„ā°æ—¶è—ā€, - èçèjCEā°Žéèšèç† SysTick æ°çš°é°æ°çžā€, -
ā½°“ā%◆èç°æ²jæœ%œæŽ¥ā...¥ RTC æjæ—¶ā€◆ā,²ā°£æjæ—¶æ°ç½°ç»œæjæ—¶i¼CEæ%œ
€ā»ææ°ç”µ/āœ°ā½°ā°āžā¼šéçæ°æ°ā°ā°ā»¶æž„ā°æ—¶è—`ā¼°āš<è@jæ—¶ā€
, æœ°æ°jæž„ā°æ°ç»“æžœi¼š ``text FLASH: 27636 B / 256 KiB RAM: 4240 B / 48 KiB ``
æœ°æ°jçfšā½°ā°CEè°»āžæ jé°CEi¼š ``text Flash written and verified! jolly good! SHA256:
1a87aec4fb68ef1dbc905cebd6b89b8bd6d9a0a436996e1b60f6f06099f438fe `` ## 15. ā@žé™...
ā±◆ā¹·ā°ā°,äž@æ£ èš„ā°Ÿā°è“◆è°“è%²ç°CEé°çæ°çæ°ā½°ā°āžā±◆ā¹·ā,é—i¼CEā°ā°ā-
~āœ°ā°çš°ā™āç,¹āCEāŸŸā€,è ŸçŽ°è±jè°æ°žā¹<ā%œæCE%œ`LCD 1.14inch` é©±āš”ā°†™ā...
Ÿçš„,`240x135` æ°¼çœ°ç°ā—ā°£i¼CEā°æ°èjç°—ā°†ā@žé™...ā±◆ā¹·ā,é—`āCEāŸŸā€, ā°æ°”
KIT é©±āš”é...◆ç½°@i¼š - `LCD 1.14inch`i¼šæ°ā±◆`240x135`i¼CEāœ°ā°æ°ā°◆çš°`x +
40`ā€◆`y + 53`ā€, - `LCD 1.47inch`i¼šæ°ā±◆`320x172`i¼CEāœ°ā°æ°ā°◆çš°`x + 0`ā€◆`y +
34`ā€, ā@žé™...çŽ°è±jæ,Ž 1.47 ā°,ā±◆ā°»ā°i¼š`320x172` ā±◆ā¹·ā,æ°¼çœ°`240x135` ā°†...

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â®¼Eä¼Ÿç•™ä, <ä¼Ÿçº| 40 äƒç´ ã€äŠ, <º| 18 äƒç´ çš, ä¯¹ç§°æªä·æ-ªEªÿÿã€
, ª²ä¼®æfi¼Ÿ - `firmware/CMakeLists.txt` ä, `LCD_ROOT` ª²ª±æä` `Examples/LCD
1.47inch/Components/Devices/TFTLCD`ã€, - æº<è¼“éŹæ¼ç•Eéçä,fª±Eª²æEº
`320x172` é±æ-ªº³¼¤äªEª±...ä,ã€, - ª...ª±æ, ...çº†ª½¿ç”” `LCD_Fill(0, 0,
LCD_Width - 1, LCD_Height - 1, color)`¼Eéä... è¶Ÿç•Eä€, ä¼®æ-
£äZæž,ªºç»“«æžœi¼Ÿ ``text FLASH: 27524 B / 256 KiB RAM: 4240 B / 48 KiB `` ä¼®æ-
£äZçfšª½ªªEe»ªžæ jéEfi¼Ÿ ``text Flash written and verified! jolly good! SHA256:
6db7f399329685225b9eca20833d9154a0f4fa2689b419313520429f9de89386 ``

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